

MATH 350-2 Advanced Calculus

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Outline

- 1 Real Analysis Lecture 7
 - More on component intervals
 - Closed Sets

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Component intervals

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- $(-\infty, 0)$ is also a component interval of $\mathbb{R} \setminus \{0, 1, 2, 3\}$
- we will show all open sets of $\mathbb{R}$ are made of component intervals!

Component intervals

Theorem (Apostol 3.10)

If $U \subseteq \mathbb{R}$ is open and $x \in U$, then there is a unique component interval of U containing x .

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Claim: (a, b) is a component interval of U containing x . $\square$

Challenge

Problem

Prove that (a, b) is a component interval of U .

Open subsets of $\mathbb{R}$

Theorem (Representation Theorem for Open Sets in $\mathbb{R}$)

If $U \subseteq \mathbb{R}$ is open, then U is the union of a countable family of disjoint open intervals.

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Uh oh ... this isn't a **countable** union.



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Any interval contains a rational number $r \in I_x$.

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Thus $x \in \bigcup_{r \in U \cap \mathbb{Q}} I_r \subseteq U$.

Since x was arbitrary,

$$U = \bigcup_{r \in U \cap \mathbb{Q}} I_r$$



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Examples:

- singleton sets!
- products of closed intervals

Challenge!

Problem

Prove that a singleton set

$$A = \{\vec{a}\}$$

in $\mathbb{R}^n$ is closed.

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We must show $U = \mathbb{R}^n \setminus \{\vec{a}\}$ is open.

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Suppose $\vec{x} \in U$.

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We must show $U = \mathbb{R}^n \setminus \{\vec{a}\}$ is open.

Suppose $\vec{x} \in U$. We must show $\vec{x}$ is an interior point of U .

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Take $r = |\vec{x} - \vec{a}|$.

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Take $r = |\vec{x} - \vec{a}|$. Then $B(\vec{x}; r)$ does not contain $\vec{a}$.

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Therefore $B(\vec{x}; r) \subseteq U$, so that $\vec{x}$ is an interior point.

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Therefore $B(\vec{x}; r) \subseteq U$, so that $\vec{x}$ is an interior point.

Since $\vec{x}$ was arbitrary, this proves U is open.

Challenge!

Problem

Prove that the **closed square**

$$[a, b] \times [c, d] = \{(x, y) \in \mathbb{R}^2 : a \leq x \leq b, c \leq y \leq d\}$$

is a closed set.

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Union of open sets is open!

Adherent and Accumulation Points

Definition

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- suprema and infima are accumulation points!
- 0 is an accumulation point of $\{1/1, 1/2, 1/3, \dots\}$

Characterizing accumulation points

Theorem (Apostol Theorem 3.17)

A point $\vec{x}$ is an accumulation point of A if and only if for all $r > 0$, the ball $B(\vec{x}; r)$ contains infinitely many points of A .

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Obviously, if for all $r > 0$, the ball $B(\vec{x}; r)$ contains infinitely many points of A , then it contains at least one point of A different from $\vec{x}$, so it's an accumulation point.

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Suppose instead that $\vec{x}$ is an accumulation point and let $r > 0$. It suffices to show that the set $C = (B(\vec{x}; r) \setminus \{\vec{x}\}) \cap A$ is infinite.



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It is also nonempty and has only positive values.

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Since $\vec{x}$ is an accumulation point, $B(\vec{x}; s) \cap A$ has an element $\vec{y}$ different from $\vec{x}$.

However, $\vec{y} \in C$ and $|\vec{y} - \vec{x}| < s$, contradicting the minimality of s .

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We conclude that C is infinite. □

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Theorem (Apostol Theorem 3.18, 3.20, 3.22)

A set is closed if and only if it contains all of its adherent points (ie. $A = \bar{A}$), or equivalently if and only if A contains all of its accumulation points.

Closure of a set

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If $\vec{x} \in \mathbb{R}^n \setminus A$, then there exist $r > 0$ such that $B(\vec{x}; r) \subseteq \mathbb{R}^n \setminus A$.

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Proof.

Suppose that A is closed.

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If $\vec{x} \in \mathbb{R}^n \setminus A$, then there exist $r > 0$ such that $B(\vec{x}; r) \subseteq \mathbb{R}^n \setminus A$.

This means that $B(\vec{x}; r) \cap A = \emptyset$.

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Thus $\vec{x}$ is not an adherent point of A .

Thus every adherent point of A is an element of A .

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Conversely, suppose that A contains all of its accumulation points.

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Conversely, suppose that A contains all of its accumulation points.

If $\vec{x} \in \mathbb{R}^n \setminus A$, then $\vec{x}$ is not an accumulation point.

Therefore there exists $r > 0$ with $B(\vec{x}; r) \cap A = \emptyset$.

This implies that $B(\vec{x}; r) \subseteq \mathbb{R}^n \setminus A$.

Closure of a set

Proof.

Suppose that A is closed.

Then $\mathbb{R}^n \setminus A$ is open.

If $\vec{x} \in \mathbb{R}^n \setminus A$, then there exist $r > 0$ such that $B(\vec{x}; r) \subseteq \mathbb{R}^n \setminus A$.

This means that $B(\vec{x}; r) \cap A = \emptyset$.

Thus $\vec{x}$ is not an adherent point of A .

Thus every adherent point of A is an element of A .

Conversely, suppose that A contains all of its accumulation points.

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Since $\vec{x}$ was an arbitrary element of $\mathbb{R}^n \setminus A$, this proves $\mathbb{R}^n \setminus A$ is open.

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Closures are closed

Theorem

If $A \subseteq \mathbb{R}^n$ is any set, then $\overline{A}$ is closed.

Proof.

Exercise! □

Closed balls are closed

Problem

Show that the closed ball

$$\overline{B}(\vec{x}; r) = \{\vec{y} \in \mathbb{R}^n : |\vec{x} - \vec{y}| \leq r\}$$

is a closed set.